

Lab 3 2024 Assembler report

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Coding approach

- I approached the problem in a stage wise manner as provided in the guidelines.
- I have created separate c files for separate type of instructions with functions that parse and encode to have an organise, easy to debug and update code.
- I included these separate files in the main file assembler.c.
- I have extensively made use of structs in parsing in order to enhance the readability of the code.

Issues encountered and their resolution

- Firstly, creating an algorithm to parse instructions was challenging, as different instructions need to be handled separately.
The solution was making use of structs and keeping them in separate code as mentioned in previous section.
- Next, handling erroneous inputs seemed complicated at first. I resolved the issue by storing binary code of all lines in a dynamically allocated array till the whole file is covered and if any error was detected, the hex file will not be written and the program would exit showing the line number and the contents of the line.
- After that, handling branch and jump instructions were difficult due to the need of having to detect labels and calculate the offset properly. I reorganised the main assembler file to store all labels in a separate dynamically allocated array and decode the branch and jal commands at the end.
This approach allowed me to report missing label errors and do parsing in a more effective manner

- Out of personal interest, I chose to handle comments and empty lines and faces numerous corner cases, including one where I found no effective solution and hence had to place a restriction on size of comments to not be exact 48 (this is due to null character being present by default there)
- Finally, debugging took up a lot of effort and I had to check and recheck with specifically crafted corner cases

Debugging approach

- First, I tested general codes taken from various sources (including the ones that sir included in the project specification.)
- Next, I came up with multiple corner cases with respect to limits on immediate/offset, register names, labels, etc.
- I stored the test cases and re used them whenever I completed a stage to ensure previous functionality was not broken.

Limitations

- Does not handle cases where spacing between different parts of command are not as per project specification
- Does not support instructions not given in project specification
- Quality of code is poor with repeated lines
- Does not give all errors at once
- Does not support .data
- The algorithms used are not efficient
- The error messages does not inform what kind of error is there, it only prints the erroneous line